

Problem Set 3: Making Money with ML?

“It’s all about location location location!!!”

1 Introduction

A new start-up dedicated to buying and selling properties has hired you and your team to develop a predictive model. Their goal is to buy as many properties as possible in the Chapinero neighborhood of Bogotá, Colombia, while spending as little as possible.

The company has access to property-level data from Bogotá sourced from [Properati](#). However, information specific to Chapinero is largely missing. Your model has to travel from one part of the city to another, and that is part of the challenge.

Critically, the company is determined to avoid a repeat of Zillow’s costly mistake.¹ Zillow developed algorithms to automate home buying, but their models systematically overestimated property values. As a result, the company lost nearly USD 500 million and laid off approximately 25% of its workforce. The lesson is that in this business, not all errors are created equal: an overprediction destroys capital, an underprediction at worst means a missed deal.

This tension between predictive performance and business viability is at the heart of this problem set:

What is the best model to predict housing prices in Chapinero, and can you trust it enough to buy?

This question should inform your choices from the very first stage.

2 General Guidelines

2.1 Deliverables

Each team must submit the following materials:

- **Slides for in-class presentation.** Each team must prepare **two** separate slide decks. Each slide deck should be submitted as a `.pdf` file and uploaded to the corresponding activity in Bloque Neón before class. Files must follow the naming convention: `nombre_equipo_XX.pdf`, where `nombre` is `best` or `compare`, and `XX` is the team number (use a leading zero for team numbers below 10). For example, for Team 1 the files would be:
 - `best_equipo_01` (Best Model Deep Dive)
 - `compare_equipo_01` (Algorithm Comparison)

¹For more on this case, see: [Flip Flop: Why Zillow’s Algorithmic Home Buying Venture Imploded](#)

- **Public GitHub repository.** Each team must submit a link on Bloque Neón to a **public** GitHub repository containing all code required to reproduce the analysis and predictions. The repository must follow the provided [template](#) and show at least five (5) substantial contributions from each team member on the main branch. Code must be fully reproducible, readable, and well-commented. I encourage you to follow the [tidyverse style guide](#). The README should help the reader navigate your repository; for example, [Project Awesome](#) has a curated list of interesting READMEs.
- **Kaggle submissions.** Predictions must be submitted on [Kaggle](#). Submissions are evaluated using the **Mean Absolute Error (MAE)**. Each team must submit at least **twenty (20) predictions** using models trained with **at least one model from each of the seven required algorithm families**² listed below: *Linear Regression, Elastic Net, CARTs, Random Forest, Boosting, Neural Networks, and SuperLearners*.

2.2 Presentation Format

Each team will deliver a **15-minute in-class presentation**. On the day of the presentation, each team will be **randomly assigned** which of the two decks to present. All team members must be equally prepared to present either deck.

Presentations should clearly communicate the empirical analysis to an audience of economists. Each slide deck should tell a coherent empirical story. Focus on interpretation and conclusions rather than mechanical reporting. Tables and figures should be self-contained, properly formatted, and publication-quality, with a title, labeled axes, and a legend. Tables must not be screenshots from R, Python, or any other software. No code in the slides. Avoid excessive text.

Golden Rule: Every slide should earn its place: if it does not advance the story the deck is telling, remove it.

2.3 Evaluation Criteria

A full rubric for the presentation and repository can be found in Bloque Neón.

3 Building the Predictive Model

To make this concrete, the price of a house can be expressed as a function of its characteristics, following Rosen's foundational work on hedonic prices:³

$$P_i = f(c_{i1}, c_{i2}, \dots, c_{in}) + u_i$$

²Kaggle imposes a daily limit on the number of submissions. Plan accordingly and do not leave submissions for the last day.

³Rosen, S. (1974). "Hedonic Prices and Implicit Markets: Product Differentiation in Pure Competition". *Journal of Political Economy*, 82(1), 34–55.

where the c_{ij} are structural attributes (e.g., size, number of bedrooms), neighborhood public services (e.g., school quality, transit access), and local amenities (e.g., parks, security, air quality). Theory tells us that price depends on these characteristics, but it does not tell us the functional form of $f(\cdot)$. That is what you have to learn from the data.

Building a predictive model is an iterative process. You will train models, submit predictions to Kaggle, receive feedback from the public leaderboard, and use that feedback to refine your approach: adjusting features, tuning hyperparameters, or trying new algorithms. Each iteration should be reflected as a meaningful contribution in your GitHub repository. The stages below describe the components of this process; the slide decks present the final result of this iterative process.

The predictive modeling process can be organized into five stages.

3.1 Stage 1: Understanding and Expanding the Data

The data comes from [Properati](#) and contains property-level information for Bogotá. The training data covers properties across the city, while the testing data covers Chapinero. This is by design: the company wants to buy in Chapinero, but most of the available data lives elsewhere. Your model has to travel from one to the other.

The Properati data alone is not enough. You are required to expand it with at least **eight additional variables**:

- At least **four predictors from external sources** (e.g., OpenStreetMap). Distances to amenities, density of services, accessibility to transit, anything that captures the local context a buyer would care about.
- At least **four predictors built from the title or description** of the listings. The free-text fields contain information that the structured columns miss. Extracting it well is part of the job.

With this expanded dataset in hand, the first task is to understand it. Examine the distribution of prices, the structure of missing values, and the relationship between candidate predictors and the outcome. Use your knowledge as economists to guide your exploration. Following Rosen, what structural attributes, public services, and amenities should drive prices in Bogotá? What patterns in the data confirm or challenge those priors? How does Chapinero differ from the rest of the city, and what does that imply for a model trained mostly outside of it?

3.2 Stage 2: Training and Comparing Algorithms

With a clean and expanded dataset, begin training regression models using the required algorithms listed in the deliverables. Start simple, submit to Kaggle, and use the leaderboard feedback to iterate: try different feature sets, adjust hyperparameters, and refine your approach. For each algorithm, select hyperparameters through a systematic process: define the range of values to explore, choose a search strategy (e.g., grid search, random search), and use cross-validation to guide

your choices. Once you have the best submission from each algorithm, compare them: go beyond Kaggle scores and try to understand *why* some algorithms outperform others on this task. What role do specification choices, tuning, feature engineering, and the structure of the data play in the performance differences?

3.3 Stage 3: Spatial Validation

Standard cross-validation assumes that observations are exchangeable: any property could end up in any fold, and the validation error is a clean estimate of out-of-sample performance. But in this problem the test set is not a random sample of the training set, it is a specific neighborhood, Chapinero, with its own price level, its own building stock, and its own amenities. A model that predicts well on a random Bogotá fold may fail when asked to price a Chapinero apartment.

Implement **spatial cross-validation** alongside regular cross-validation. The split can be done spatially, by UPZ, by neighborhood, or by spatial grid; the choice is yours, but justify it. Compare the two validation strategies for your candidate models: do they agree on which model is best? Which one tracks the Kaggle public leaderboard more closely? What does the gap (or the lack of it) tell you about how confident you can be that the model will work in Chapinero?

3.4 Stage 4: The Best Model

From the models trained in Stages 2 and 3, identify the one that achieved the team's highest public score on Kaggle. Now ask yourself: why did this model win? What is the algorithm doing that the others are not? Examine the hyperparameter configuration that led to the best performance. Investigate feature importance: which variables are the strongest predictors and why? How much of the predictive power comes from the original Properati variables, how much from the OSM features, and how much from the text-derived features? What is the model capturing about how prices form in Bogotá, and how well does that travel to Chapinero?

3.5 Stage 5: From Prediction to Business

Your submissions are evaluated using the **Mean Absolute Error (MAE)**. Think carefully about what this metric means in a property-buying context. The MAE treats every peso of error as equally costly, regardless of direction: an overprediction of *COP 10 million* counts the same as an underprediction of *COP 10 million*. But the business does not see it that way. **An overprediction means the company pays more than the property is worth, i.w. money lost.** An underprediction means the company offers too little and walks away from the deal, an opportunity foregone, but no capital destroyed. This is the asymmetry that cost Zillow nearly USD 500 million.

How does your best model balance these two types of errors? Does it overpredict or underpredict on average? In which segments of the market does it fail most? What does this trade-off imply for the design of a buying strategy? Would a buyer prefer a different balance than the one MAE rewards?

Beyond the model's errors, consider whether it could be deployed in practice. Going back to the central question: *can you trust this model enough to buy?* If the company gave you a budget to acquire properties in Chapinero tomorrow, would you let the model decide which ones, and at what price? What would you recommend to the company before they put real money behind the predictions?

4 Slide Decks

Each deck maps onto the stages above.

- **best_equipo_## Best Model Deep Dive.** This deck tells the story of your best model from the data up and connects it to the business problem. The first slide is mandatory (see below). Beyond that, teams are free to organize their deck, but a recommended structure is:
 - *Mandatory first slide:* A *Best Model Overview* slide with the specification of the best model and the Kaggle score (public leaderboard). State the main takeaways upfront and begin constructing the story.
 - *Data and features (2–3 slides):* Use your domain knowledge as economists to show the audience why the variables in your best model matter. Highlight the role of the external (OSM) features and the text-derived features: what economic reasoning led you to build them, and what the data revealed.
 - *Best model (3–4 slides):* Deep dive into the training process, hyperparameter tuning, feature importance, and diagnostics. How much of the predictive power comes from the original Properati variables, the OSM features, and the text-derived features?
 - *Spatial validation (1–2 slides):* Compare regular cross-validation and spatial cross-validation for your best model. What does the gap (or the lack of it) tell you about how confident you can be in Chapinero predictions?
 - *Business implications (1–2 slides):* Discuss what the MAE reveals about your best model's errors. Does the model systematically overpredict or underpredict? How does it balance the asymmetric costs of overpaying versus missing a deal? Going back to the central question: *can you trust this model enough to buy?*
- **compare_equipo_## Algorithm Comparison.** This deck positions the best model against the alternatives. The focus is on the algorithms themselves, what each one is doing, and why some win and others lose on this task. The first slide is mandatory (see below). Beyond that, teams are free to organize their deck, but a recommended structure is:
 - *Mandatory first slide:* A *Best Model Overview* slide with the specification of the best model and the Kaggle score (public leaderboard). State the main takeaways upfront and begin constructing the story.

- *Data context (1–2 slides)*: A brief description of the feature set used for the comparison. Use your understanding of the data and the problem to explain why certain algorithms and specifications may work better than others.
- *Algorithm comparison (4–5 slides)*: Present a summary of all seven required algorithm families, then focus the detailed comparison on the two algorithms that came closest to your best model's performance. A meaningful comparison requires a level playing field: differences in scores should reflect the algorithm, not differences in the set of predictors or preprocessing choices. Use the deep dive into the top contenders to explain what drives the performance gap.
- *Conclusion (1 slide)*: What the comparison reveals about the trade-offs between model complexity, interpretability, and predictive performance. If the company asked you to recommend one model for buying in Chapinero, would it necessarily be the one with the highest Kaggle score?